

```

(0) - genotyping_in_silico: 5 windows
(1) -> + 0: bash-
(2) -> + 1: tree_calc*
(3) -> + 2: bash
(4) -> + 3: bash
(5) -> + 4: bash
(6) - 1: 1 windows
(7) -> + 0: bash*
(8) - 2: 2 windows (attached)
(9) -> + 0: http-
(M-a) -> + 1: [tmux]*Z

1 (sort: index)
  x $ t 2 3 Z h < 1
  p l R q I * % R . *
  B 7 ; z C I + , ( 7 H u
  - 2 3 w T A G J f
  - U R S z 9 # [ 0 Z
  f V b j + - z U 9 D x
  d z B g ; A D P ^ 3 0
  . : z 8 W u 7 H ^ Q Y
  : U w N y v @ M K Y h
  Q @ j r b R A x > j ) W 8 1
  ] # 0 B 1 X 7 0 < \
  x I 2 G 0 [ 0 1 > 0 y + h
  < m : 2 T 1 D ) u Y $ a
  - + [ N z + y b K > 7 U # * $
  q A ' Y ( N < H b > / < E : F
  C r S I b @ 4 G - 1 D , 6
  X j ! C 0 f D 2 7 - = q ;
  - a 2 % u T : 0 1 / 5 b > /
  u Z M k 0 P k d m 8 d \ n
  X * : j ( Z B p # 7 W N b X
  T h L ( ) 5 d K / j h s m C

0[ 0.0%] 4[ 0.0%] 8[ 3.3%] 12[ 0.0%] 14[
1[ 0.0%] 5[ 0.0%] 9[ 0.0%] 13[ 0.0%] 15[
2[ 0.0%] 6[ 2.0%] 10[ 0.0%] 16[
3[ 2.6%] 7[ 0.0%] 11[ 0.0%] 17[
Mem[ ] 842M/236G Task
Swp[ ] 14.8M/1024M Load
Upt[ ]

Main 1370
PID USER PRI VIRT RES SHR S C
101546 ubuntu 20 1 5852 3792 3796 S
294595 ubuntu 20 6644 2712 2256 S
294334 ubuntu 20 0 9072 5028 3568 S
F1Help F2Setup F3Search F4Filter F5Tree F6Sort F7

[2] 0:htop- 1:[tmux]*Z
"quar2025-fa895" 01:41 11-Mar-28

```

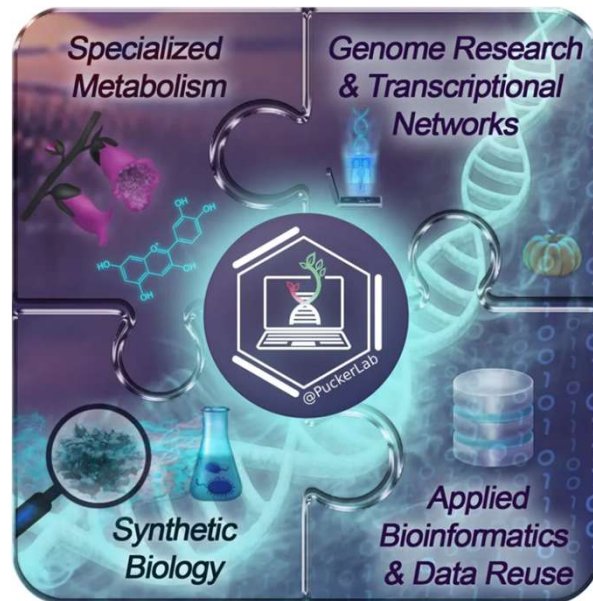
Samuel N. Meckoni & Nancy Choudhary

# Command-line interface (CLI) basics for life scientists

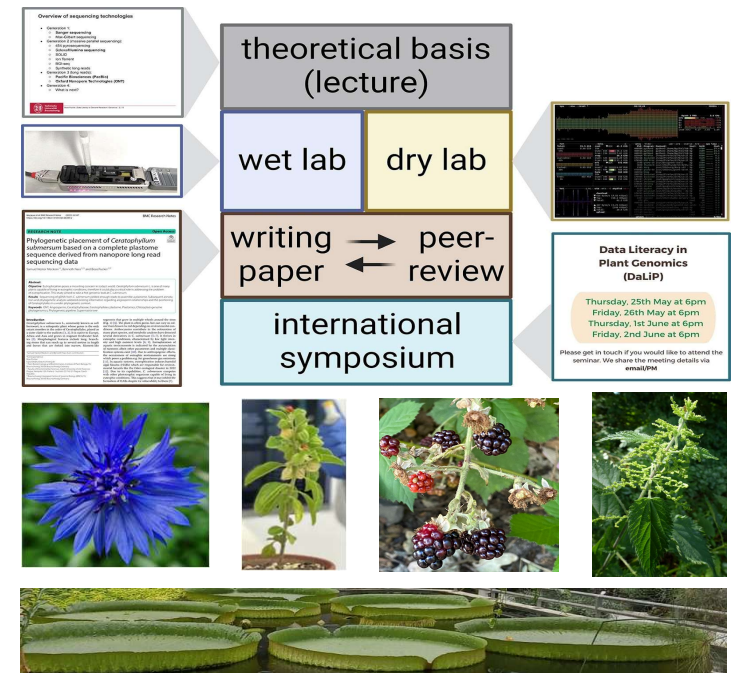
# Plant Biotechnology & Bioinformatics (PBB)

## Group overview

- Flavonoids:  
Anthocyanins,  
Flavonols, Flavones,  
Proanthocyanidins
- Carotenoids
- Withanolides
- ...



Tools: KIPES, NAVIP, MGSE,  
MYB annotator, DuplyliCate...



Wolff *et al.*, 2023: 10.1515/jib-2023-0033 2

Dassow *et al.*, 2025: 10.1101/2025.01.31.635922



# Course Overview

## CLI basics for life scientists

- Basics of the Unix command line interface
- Unix file system
- Manipulating files and directories
- Combining bash commands
- Using these commands for biological insights



Samuel



Nancy



Boas

# Availability of course material

- All materials will be shared at the end of this course.

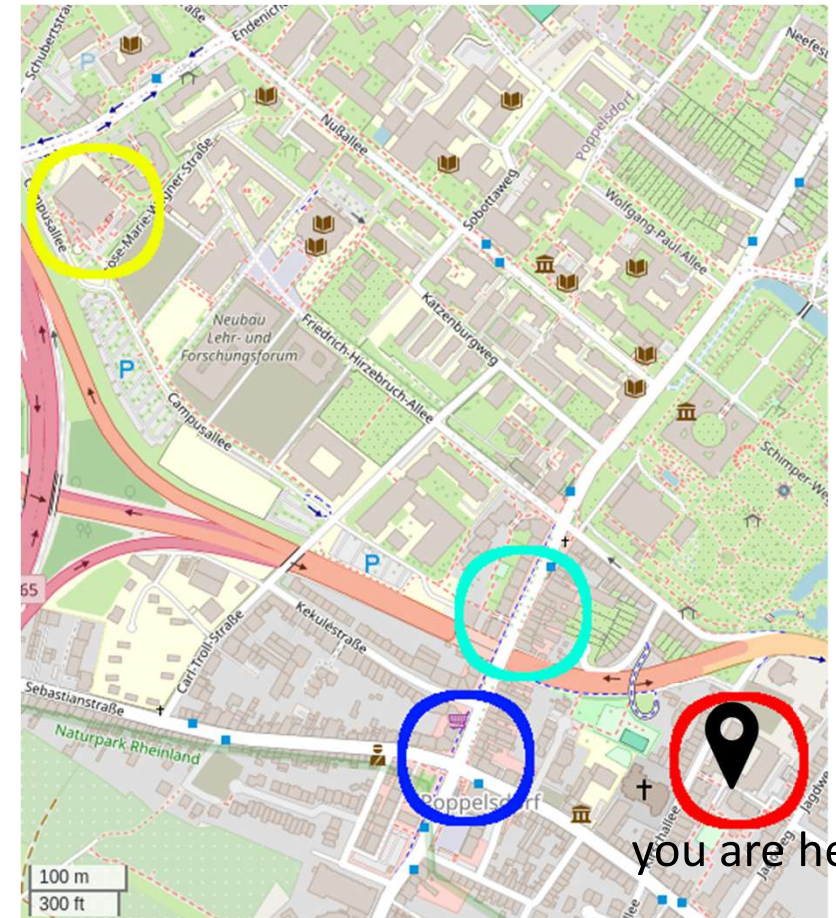
[https://github.com/PuckerLab/teaching/tree/main/deNBI2026\\_CLIbasicsforlifescientists](https://github.com/PuckerLab/teaching/tree/main/deNBI2026_CLIbasicsforlifescientists)

# Organisational I

- 10 AM -5 PM, 1 hour lunch break
- Tuesday: lunch break from 1 PM - 3 PM
  - PhD defense at 2 PM in 2.007
- Friday until 1 PM

# Organisational II

- Mensa (15 mins walk)
- Local food stores
- Supermarkets



you are here

# Recap

- Using CLI commands for gaining biological insights
- How to prepare data for transcriptomics analysis

# Long runs

- Some runs will take a long time to finish (hours < days < weeks)
- ssh connections usually don't last as long → the run will be interrupted
- Solution: background process

# Background and foreground

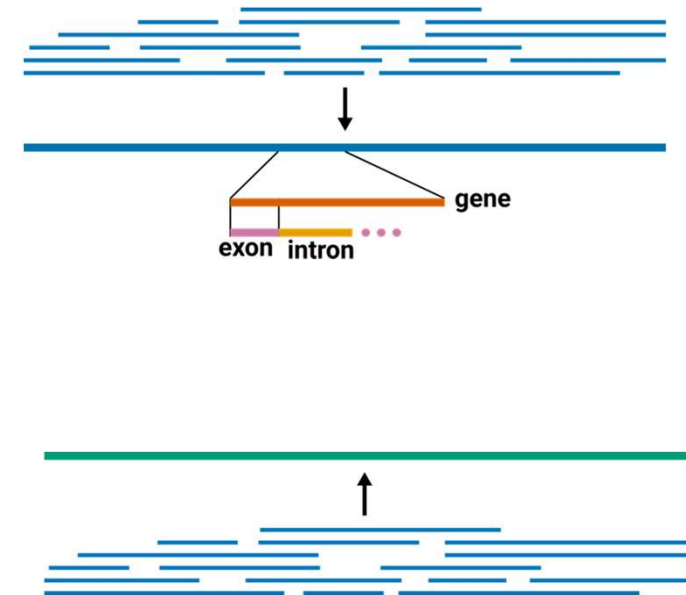
- Ctrl (/Strg) + Z to stop the command
- fg = bring job to foreground
- bg = bring job to background
- or append a "&" at the end of the command before running it

# Alternative : tmux

- Multiplexer
- Multiple CLI windows and panes in one session
- Keeps running in the background, even if you stop the ssh connection

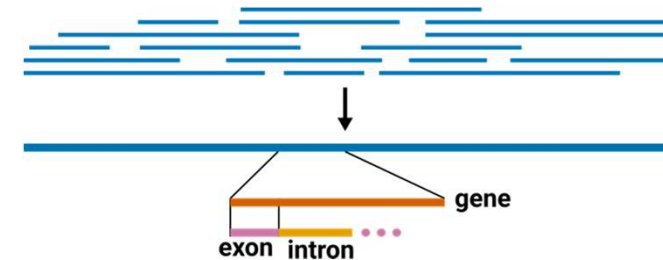
# A typical genomics workflow

- Assembly
- Annotation
- Read mapping



# A typical genomics workflow

- Assembly
- Annotation
- Read mapping

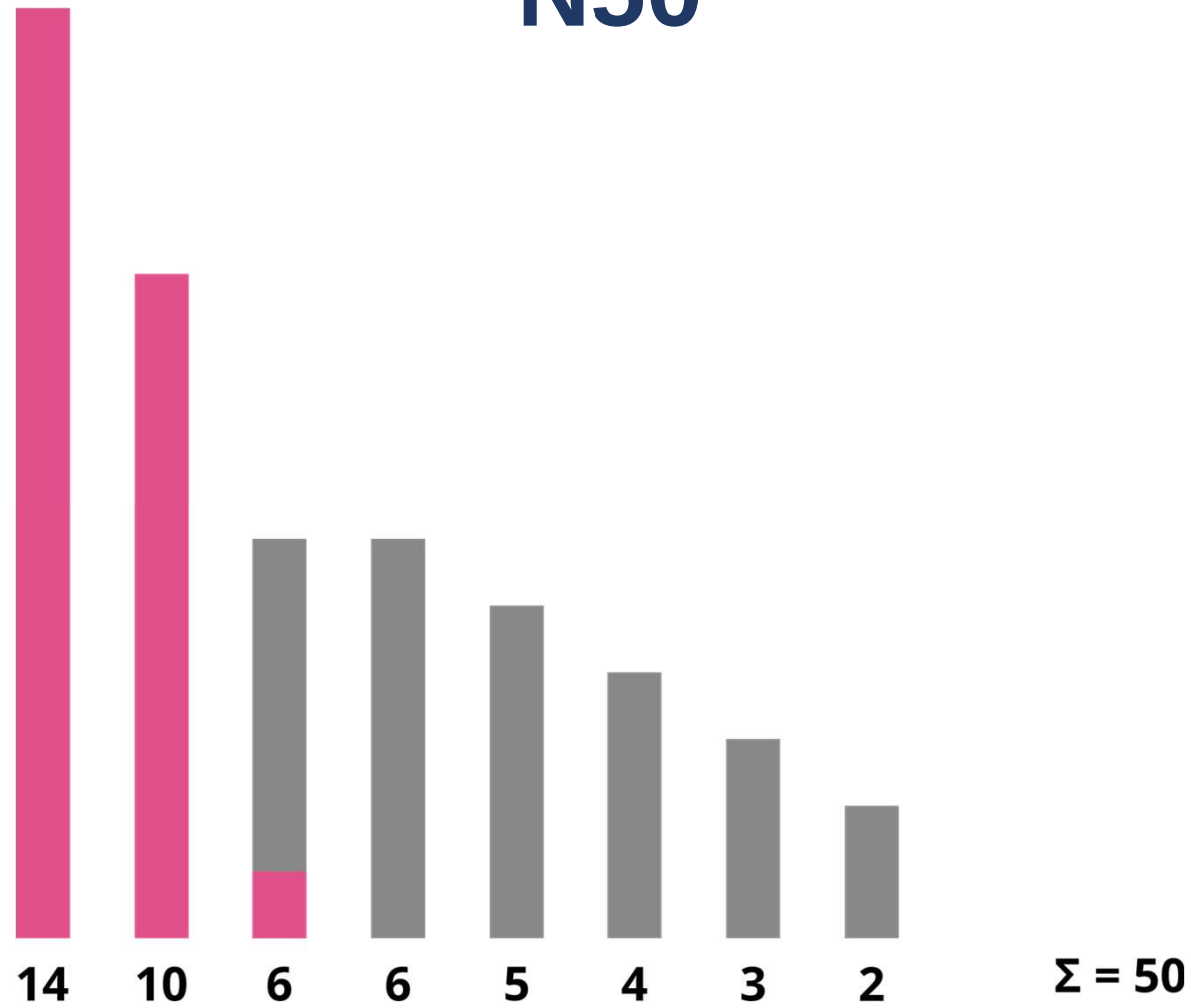


# Metrics

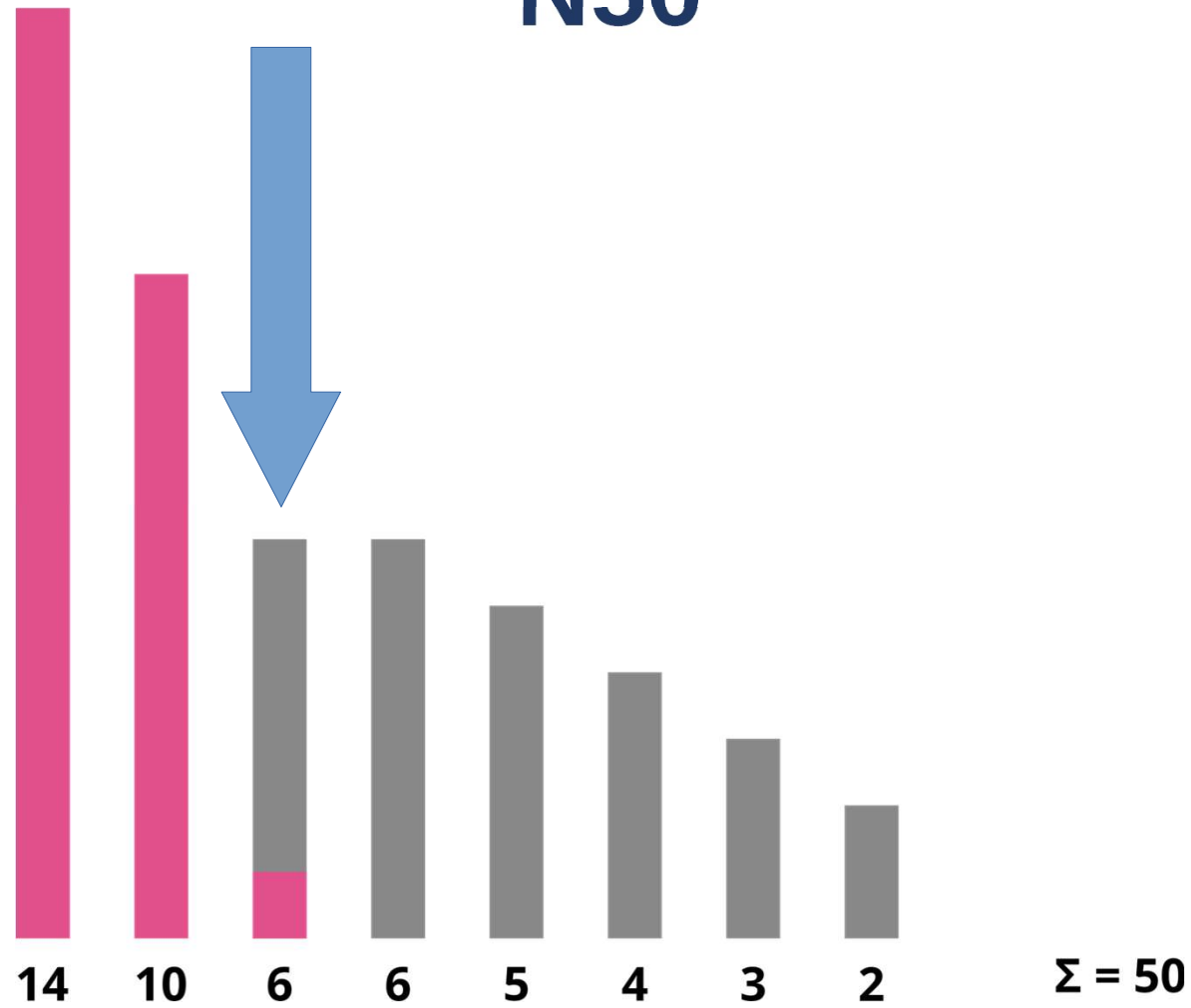
- Amount of reads
- Coverage depth



# N50



**N50**



# Exercises

- Task 3.1: Download additional data from the shared VM:
  - Genome contig:  
`/vol/data/task3/daUtrGibb.v01.asm.ptg000007l.fasta`
  - Reads subset:  
`/vol/data/task3/subsample_u_gibba.0.01.2.fastq`

# Exercises

- Task 3.2: install seqkit with apt and get some statistics from a fastq and a fasta file from the previous day:
  - No. of sequences
  - Longest sequence
  - How many sequences are longer than 1 Mbp
  - Can you get the N50?

# Exercises

- Task 3.3: Use minimap2 to map subsample\_u\_gibba.0.01.fastq against daUtrGibb.v01.asm.ptg000001l.fasta
- Task 3.4: Use samtools to sort and index file resulting in an alignment file

# Exercises

- Task 3.3: Use minimap2 to map subsample\_u\_gibba.0.01.fastq against daUtrGibb.v01.asm.ptg000001l.fasta
- Task 3.4: Use samtools to sort and index file resulting in an alignment file
- Task 3.5: Redo task 3.3 and 3.4, but this time in one command

# Exercises

- Task 3.6: Calculate the read coverage with samtools
  - Which position has the highest and which has the lowest coverage?
  - What is the average coverage (arithmetic mean and median)?

# Scripts

# How to execute multiple commands?

- scripts
- write multiple commands into a file
- “execute” the file once
- execute like this
  - `bash /path/to/script.sh`
  - `/path/to/script.sh`
  - `source /path/to/script.sh`
  - ...

# Loops

- **while loop:**  
while read line; do  
    echo \$line  
done < ./ids
- **for loop:**  
for i in {1..3}; do  
    echo "Hi"  
done



# First small script

- Task 4.1: write a script that creates a new directory. In this directory, it should create a file that contains 10 times the output of the “fortune” command. Then it should print out this file.

# Variables

- variables ...
- define a variable  
var1="text"  
var2=0
- program output as variable:  
var3=\$(fortune)
- call a variable with "\$" and the name of the variable  
e.g.: echo \$var2

→ Also works in the CLI



# Positional arguments/variables

- Can be called in the script with \$1 or \$2, \$3, ...
- Equals the positional input arguments
- Example:

```
sam@debian-nds: ~  
(base) sam@debian-nds:~$ cat echo_script.sh  
echo $2 was given after $1  
(base) sam@debian-nds:~$ bash echo_script.sh 2 hi  
hi was given after 2  
(base) sam@debian-nds:~$
```

# Tmp files

- sometimes you want to write some intermediary output to a file that your script will use later, but you do not need it after the script finishes and the file can be deleted: a temporary file
- command “mktemp”: creates and returns the path to a temporary file
  - also temporary directories are possible

# Task: create a more useful script

- do not use LLMs
- you will present your script tomorrow and should be able to explain everything
- additional data in shared VM

# Script: Task1

- do not use LLMs
- Create a script that takes two BUSCO result tables as input and outputs the total number of BUSCO genes, missing in one file, but present in the other, as well as a list of these genes.
- Background: comparing two genome sequences/annotations based on more thorough BUSCO metrics.
- Extra: Create a similar script that has the same input, but calculates and compares the amount of duplicated BUSCO genes, including the rate of duplication per duplicated BUSCO gene and compares that between the two input files. The output should give an estimation about which file might be inferred from the "better" source.

## Script: Task2

- Create a script that takes as input all genes of a genome sequences (peptide or CDS fasta file) and a reference gene set. It should use blast to find genes of the whole genome that match the reference genes and outputs them as a new, separate FASTA file.
- Background: Finding genes of interest in a new plant species via a homology-based approach

## Script: Task3 and Task4

- Create a script that downloads chloroplast genomes from NCBI (NCBI efetch) using a list of IDs as input and calculates the average length (arithmetic and median) of these sequences.
- Like Task3 but with whole genome sequences from NCBI (NCBI datasets tool)
- Background: Gaining insights into the distribution of chloroplast and whole genome sizes.

# Script: Task5 and Task6

- Create a script that downloads protein sequences from the Uniprot database (Uniprot API queries) using a list of IDs as input and builds phylogenetic trees using them (use a multiple sequence alignment program like MAFFT and a phylogenetic tree inference program like IQTREE3).
- Like Task5 but with NCBI gene IDs
- Background: Phylogenetic trees are useful for various analyses. The construction will be a lot easier and faster when directly using database IDs.
- Extra: Extend the script to additionally incorporate a user-defined FASTA file



# Script: Task7

- Create a script that downloads raw long reads from the SRA (SRA toolkit) using a list of SRA run IDs as input and calculates average read length metrics (arithmetic median and N50). Compare these values across all samples (calculate again some averages) and also the total amount of sequenced bases per sample.
- Background: Gaining insights into the length distribution of long reads and the (potential) correlation of it to the total amount of sequenced bases per sample.
- Extra: Plot the results (e.g. gnuplot)

